

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Phase I, open-label, multi-center dose escalation study of FAZ053 as single agent and in combination with PDR001 in adult patients with advanced malignancies **Short Title/Acronym:** FAZ053 / PDR001 Phase I Study **Protocol Number:** CFAZ053X2101 (EudraCT: 2016-001470-15) **NCT Number:** NCT02936102 **Sponsor Name:** Novartis Pharmaceuticals Corporation **Investigational Product:** FAZ053 (anti-PD-L1 monoclonal antibody) and PDR001 (Spartalizumab, anti-PD-1) **Phase of Development:** Phase I **Medical Monitor:** Novartis Clinical Operations Lead **Publication Reference:** Janku F, Tan DSP, Martin-Liberal J, et al. (2025) First-in-human study of FAZ053, an anti-programmed death-ligand 1 (anti-PD-L1) monoclonal antibody, alone and in combination with spartalizumab, in patients with advanced malignancies. ESMO Open. 10(6):105051.

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To characterize the safety and tolerability of FAZ053 as a single agent and in combination with PDR001 (spartalizumab) and to identify recommended doses and schedules for future studies. * **Secondary Objectives:** To characterize the pharmacokinetic (PK) and pharmacodynamic (PD) profiles, assess the emergence of anti-FAZ053 and anti-PDR001 antibodies, characterize changes in the immune infiltrate in tumors, and assess preliminary anti-tumor activity.

2.2 Background & Rationale FAZ053 triggers an antitumor response by targeting programmed death-ligand 1 (PD-L1), thereby activating effector T cells and negatively regulating T cells. This study was conducted to evaluate whether FAZ053—administered alone or in combination with the anti-PD-1 inhibitor spartalizumab (PDR001)—can safely achieve immune gene expression and preliminary efficacy in patients with advanced solid tumors with unmet medical needs.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional (Dose Escalation and Dose Expansion) * **Control Type:** Single-arm / Non-randomized * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target** **\$N\$**: 154 patients treated * **Number of Sites**: 12 centers across 9 countries (USA [48 participants], Singapore [27 participants], Canada [23 participants], Japan [17 participants], Spain [15 participants], Israel [14 participants], Italy [6 participants], France [3 participants], Taiwan [1 participant]). * **Study Initiation Date**: 20-Oct-2016 (First participant first visit). * **Early Termination Date**: 09-Jul-2018 (Recruitment halt; dose expansion of combination therapy was not initiated due to business reasons). * **Primary Completion**: 22-Nov-2024 (Last participant last visit).

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients with advanced malignancies (solid tumors). 2. Patients previously exposed to anti-PD-1/PD-L1 treatment who are adequately treated for skin rash or with replacement therapy for endocrinopathies were eligible to participate.

4.2 Exclusion Criteria 1. Treatment with cytotoxic or targeted antineoplastics within 3 weeks of initiation of study treatment (6 weeks required for agents with delayed toxicity like nitrosoureas, mitomycin C, or prior immunotherapies). 2. Presence of symptomatic central nervous system (CNS) metastases or CNS metastases requiring local CNS-directed therapy or increasing doses of corticosteroids within the prior 2 weeks. Treated brain metastases required 4 weeks of neurological stability and patients must be off steroids for at least 2 weeks before study treatment. 3. Active, known, or suspected autoimmune disease, or a history of severe hypersensitivity to study treatment excipients or other monoclonal antibodies.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency**: FAZ053 monotherapy (e.g., 1200 mg Q3W or 1600 mg Q6W) and combination therapy with PDR001 (e.g., 300 mg Q3W). * **Route of Administration**: Intravenous (IV) Infusion * **Duration of Treatment**: Until disease progression or unacceptable toxicity. The median duration of exposure was 105 days for monotherapy and 85 days for combination.

5.2 Concomitant & Prohibited Therapy * **Permitted**: Replacement therapy for endocrinopathies and therapies for adequately treated skin rash. * **Prohibited**: Cytotoxic or targeted antineoplastics within the specific 3-to-6 week washout periods, systemic corticosteroids in escalating doses prior to trial, and concurrent immunosuppressive therapies for active autoimmune disease.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans, CNS Stability Check
Baseline	Day 0	First IV Infusion, PK/PD Sampling, Safety Labs
Interim	Q3W / Q6W	Safety Labs, Efficacy Assessments (Tumor Scans), DLT Check, Infusions
End of Study	Follow-up	Final Vital Signs, Exit Assessments, Survival/Safety Monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * Definition: Incidence of Dose-Limiting Toxicities (DLTs) and Treatment-Related Adverse Events (TRAEs) to determine the recommended doses and schedules. *** Measurement Method:** Clinical investigator assessments utilizing CTCAE criteria (e.g., monitoring for events such as Grade ≥ 3 liver function test increases or Grade 4 creatinine increases).

7.2 Secondary & Exploratory Endpoints * Pharmacokinetics: (C_{max} , AUC, terminal half-life) of FAZ053 (e.g., $ST_{1/2}$ of 20.6 days at 1200 mg Q3W). *** Overall Response Rate (ORR) - Dose Escalation:** 4.8% (Partial Response [PR]: n=2) for FAZ053 single agent Q3W, 5.9% (PR: n=1) for FAZ053 single agent Q6W, and 4.9% (PR: n=3) for FAZ053 Q3W in combination with PDR001 300 mg Q3W. *** Overall Response Rate (ORR) - Dose Expansion:** In specific expansions with FAZ053 1200 mg Q3W, ORR was 10.0% (PR: n=2) for TNBC, 50% (PR: n=2) for advanced alveolar soft part sarcoma (ASPS), and 30% (Complete Response [CR]: n=1, PR: n=2) for chordoma. *** Biomarker/Pharmacodynamic analysis:** Changes in immune gene expression and tumor immune infiltrate.

8. Safety & Adverse Event Reporting

- Adverse Event (AE) Monitoring:** Continuous evaluation for general and immune-mediated AEs. The most frequent TRAEs were fatigue, diarrhea, and nausea.
 - Serious Adverse Events (SAEs):** Grade ≥ 3 TRAEs occurred in 6.5% of monotherapy and 4.9% of combination therapy patients. DLTs included Grade 3 LFT increases and Grade 4 creatinine increases.
 - Data Monitoring Committee (DMC):** Regular interval reviews conducted during the dose-escalation phase to dictate progression to subsequent cohorts.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set included all patients receiving at least one dose of FAZ053 alone or with PDR001 (\$N = 154\$).
 - **Sample Size Justification:** Utilized standard dose-escalation cohort rules to identify the maximum tolerated dose (MTD) / recommended phase 2 dose, followed by planned expansion cohorts (TNBC = 20, Chordoma/ASPS = 15) to explore preliminary efficacy signals.
 - **Handling of Missing Data:** Standard oncology phase 1 censoring for patients lost to follow-up or withdrawing consent before progression.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Sponsor-led clinical monitoring (onsite and remote) in alignment with multi-national trial regulations.
 - **Audit Trail:** Centralized database capturing source-verified PK, PD, and imaging/response evaluation data.
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11. Study Identifiers & Governance

- **Primary ID:** CFAZ053X2101
 - **Registry Number:** NCT02936102
 - **Sponsor/Lead Organization:** Novartis Pharmaceuticals Corporation
 - **Study Title:** A Phase I, open-label, multi-center dose escalation study of FAZ053 as single agent and in combination with PDR001 in adult patients with advanced malignancies
 - **Official Regulatory Title:** A Phase I, open-label, multi-center dose escalation study of FAZ053 as single agent and in combination with PDR001 in adult patients with advanced malignancies
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12. Clinical Framework (Oncology Specific)

- **Primary Condition:** Advanced Solid Tumors / Malignancies (including ASPS, Chordoma, TNBC)
 - **Diagnosis Threshold:** Radiographically measurable advanced/metastatic disease unsuitable for standard curative measures.
 - **Medical Methodology:** Immuno-oncology; concurrent PD-L1 and PD-1 receptor blockade.
 - **Optimization Target:** Identifying the optimal biological dose and maximum tolerated dose (MTD).
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13. Study Procedures & Methodology

- **Management Pathway:** Step-wise dose escalation guided by DLT evaluations followed by specific solid tumor dose expansions.
- **Education Protocol (HCP):** Safety run-in training focused on the management of immune-related Adverse Events (irAEs).
- **Education Protocol (Patient):** Strict protocols for reporting immune-mediated symptoms (e.g., fatigue, gastrointestinal distress) during the DLT observation window.
- **Quality Audit Mechanism:** Continuous safety reviews between cohorts prior to approving escalation.

14. Observation & Follow-Up Schedule

- **Total Duration:** 2016 to 2024 (Approx. 8 years total study runtime).
- **Onsite Visit Cadence:** Correlated with the Q3W (every 3 weeks) and Q6W (every 6 weeks) intravenous dosing schedule.
- **Remote Monitoring Frequency:** Between treatment cycles for ongoing safety/AE tracking.
- **Checklist Review Interval:** Prior to every infusion cycle and at the end of the DLT evaluation period.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Insert Name]	_____	03-Apr-2026				
Clinical Operations Lead	[Insert Name]	_____	03-Apr-2026				
Lead Statistician	[Insert Name]	_____	03-Apr-2026				